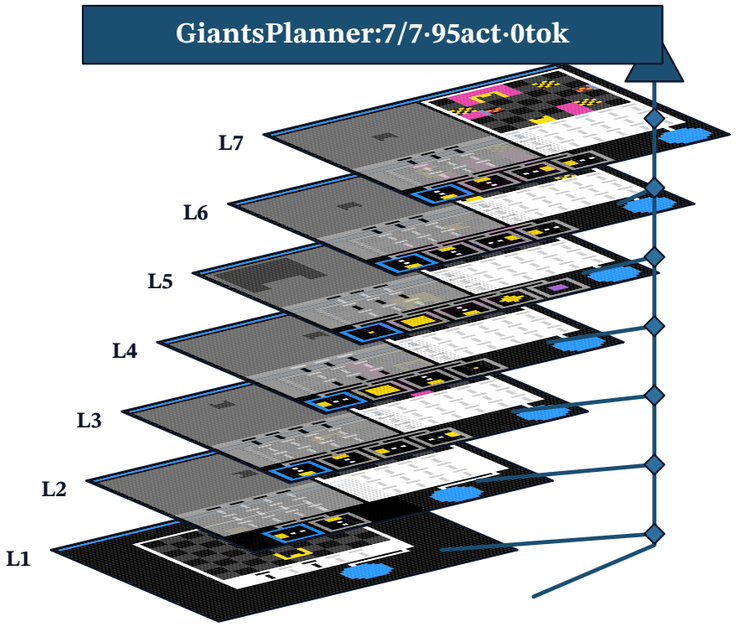
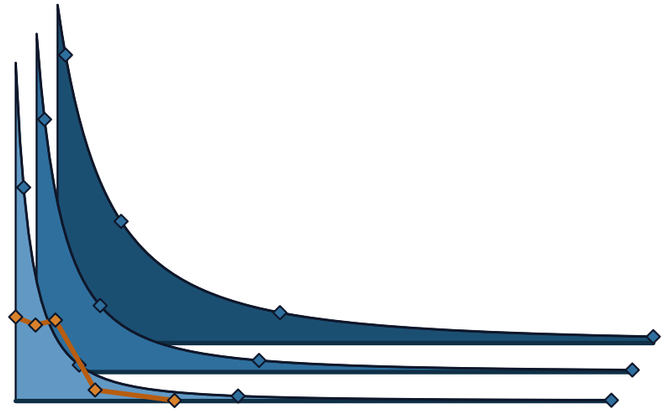
## a Frontier Lock & quadratic dilution



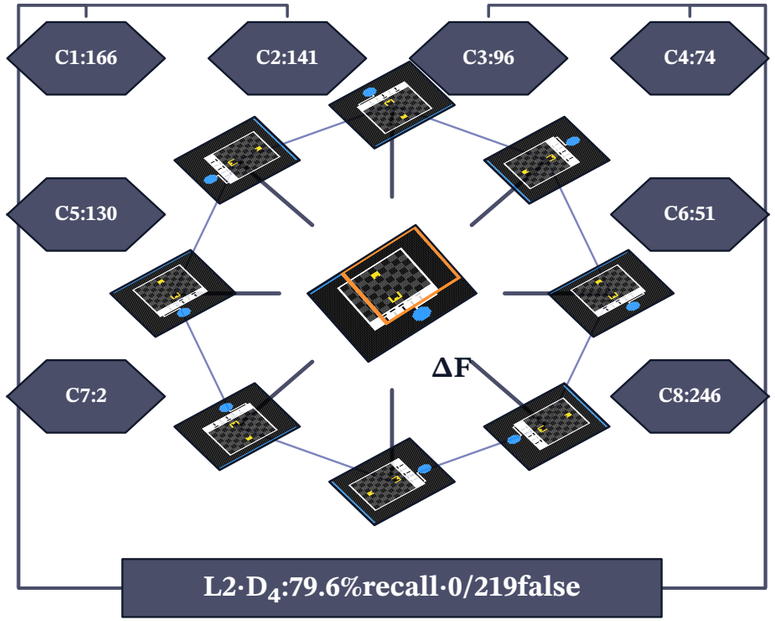

People:317act · 33runs:L1–L2 · h(80) ≤ 0.091

$$(b_l/(A+k+1))^2$$

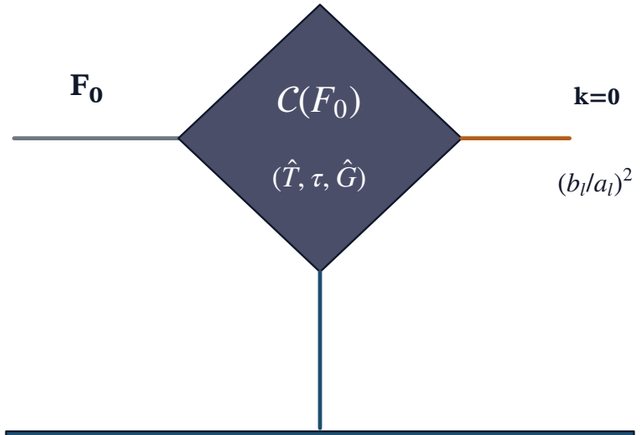


A=30/50/95 · k:0 → 600

## b D4 equivariant world model & certificate gate



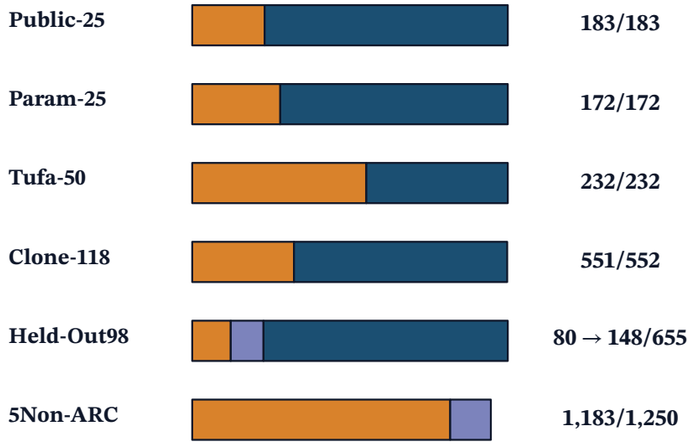

L2·$D_4$:79.6%recall·0/219false




Certified:183/183·0.413×·0tok

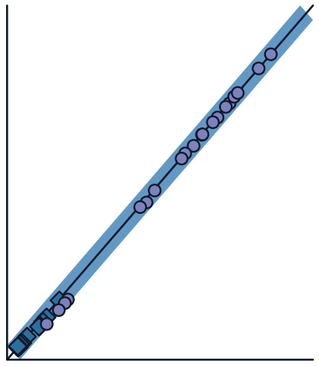
## c Generalization across 316 games & 30 Hidden-110 runs

1,793/1,794·99.94%




Baseline / +OnlineInducer / Full·C1..C8

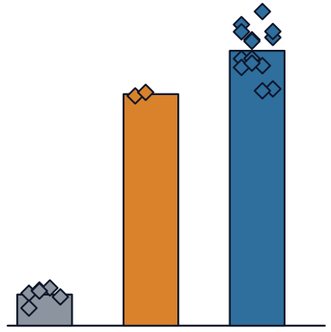
$R^2$=0.978/0.997



Predicted → Observed

30·Hidden-110

25.70·U=0·p=0.0083



vLLM / Probe / k=0

2.81 / 20.76·21.08 / 24.83±2.12